Reproducing NeuMF on MovieLens-100K

ECE480 — Deep Learning and its Applications

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Abstract

We reproduce the *Neural Matrix Factorization* (NeuMF) method of He et al. [1] on the MovieLens-100K dataset (100,000 ratings, 943 users, 1,682 items). We examine the effect of MLP depth (1–3 hidden layers) with and without pretraining, the effect of the negative-sampling ratio (1–10) and the top-K cutoff (1–10); we compare NeuMF to a non-negative matrix factorisation (NMF) baseline with k latent factors swept over $\{1, 6, 11, 16, 21, 26\}$; and we train three smaller NeuMF student models using response-based, feature-based (FitNets) and relation-based (RKD) knowledge distillation. Every experiment is repeated **10 times** and reported as mean $\pm$ standard deviation. NeuMF reaches HR@10 $= 0.649 \pm 0.011$ and NDCG@10 $= 0.390 \pm 0.007$ at the best setting; response-based knowledge distillation preserves (and slightly improves) the teacher's HR@10 while using 50.7 % fewer parameters.

1 Experimental setup (Task 1)

Dataset. The MovieLens-100K dataset (`u.data`) contains 100,000 explicit ratings by 943 users on 1,682 items; every user has rated at least 20 items. Following the NCF paper, we treat ratings as *implicit feedback*: every observed rating is a positive interaction regardless of its numerical value, and absence of an interaction is treated as a negative. The timestamp column is not used as a feature, but we do use it to order the per-user interactions for the leave-one-out split.

Dataset split (leave-one-out). For every user we sort the interactions chronologically and reserve the most recent interaction for testing, the second-most-recent for validation, and the remaining $\sim$ 98,114 interactions for training. At evaluation time we score the held-out positive item against 99 items that the user has never interacted with (these 99 negatives are sampled once per user with a fixed RNG seed, so all experiments evaluate on exactly the same candidate lists). HR@K and NDCG@K are then computed on the resulting ranked list of 100 items and averaged over the 943 users.

Parameter setting. We use the configuration recommended by [1] and validated in pilot runs on MovieLens-100K: GMF embedding dimension = 8, MLP embedding dimension = 32 per branch, 3 MLP hidden layers with a tower structure ($64 \to 32 \to 16 \to 8$), ReLU activations, batch size 256, 4 sampled negatives per observed positive, and 20 training epochs. When NeuMF is trained from scratch we use Adam with a learning rate of 10^{-3}; when it is initialised from pretrained GMF and MLP checkpoints we switch to vanilla SGD with the same learning rate, as prescribed in Section 3.4.1 of the paper, and mix the two output heads with $\alpha = 0.5$. Every experiment is repeated 10 times with seeds $0, \ldots, 9$ and we report mean $\pm$ sample standard deviation.

2 Effect of MLP depth and pretraining (Tasks 2 & 3)

HR@10 vs. number of MLP layers (Task 2). Table 1 and Figure 1 summarise the effect of the MLP depth on NeuMF's HR@10 and NDCG@10, both with and without GMF/MLP pretraining. Three observations stand out:

- **Depth brings no benefit at this dataset size.** Without pretraining, HR@10 is essentially flat from 1 to 2 layers (0.660 $\rightarrow$ 0.659) and drops slightly at 3 layers (0.649). NDCG@10 peaks at 2 layers (0.394) but the differences between depths are within one standard deviation of each other. This is the opposite of the paper's finding on MovieLens-1M, where stacking more non-linear layers monotonically improved performance. A plausible explanation is dataset size: MovieLens-100K has ten times fewer interactions than MovieLens-1M, so the expressive-capacity gain of deeper MLPs is outweighed by the increased risk of overfitting on the smaller training set.
- **Pretraining has a marginal effect on HR@10 and a slightly negative effect on NDCG@10.** With pretraining, HR@10 stays within ± 0.01 of the no-pretrain version at every depth, while NDCG@10 is consistently lower by $\sim$0.01–0.014. The most likely cause is the optimiser switch (Adam during pretraining, then vanilla SGD for the fused NeuMF): SGD without momentum requires many more updates to escape the initial basin, and our 20-epoch budget on the fused model is not enough for SGD to catch up with what Adam reaches end-to-end.
- **Pretraining does reduce variance.** The standard deviations with pretraining (0.005–0.009) are roughly half of those without (0.010–0.013), which is an expected consequence of starting from pre-converged embeddings rather than random ones.

Based on these results we adopt **3 MLP layers, no pretraining**, as the "best NeuMF setting" throughout the rest of the report. We choose depth 3 rather than 1 or 2 for fair comparison with the paper's reference architecture, even though it is not the strict HR@10 peak here; it is within one standard deviation of all other depths and matches Section 4.1 of the paper.

Table 1: NeuMF HR@10 and NDCG@10 vs. MLP depth (mean $\pm$ std, 10 seeds).

Pretraining	MLP-1	MLP-2	MLP-3
HR@10 (without)	0.6600 $\pm$ 0.0106	0.6585 $\pm$ 0.0129	0.6491 $\pm$ 0.0111
HR@10 (with)	0.6540 $\pm$ 0.0057	0.6592 $\pm$ 0.0055	0.6504 $\pm$ 0.0089
NDCG@10 (without)	0.3908 $\pm$ 0.0060	0.3942 $\pm$ 0.0054	0.3902 $\pm$ 0.0072
NDCG@10 (with)	0.3771 $\pm$ 0.0034	0.3861 $\pm$ 0.0047	0.3848 $\pm$ 0.0058

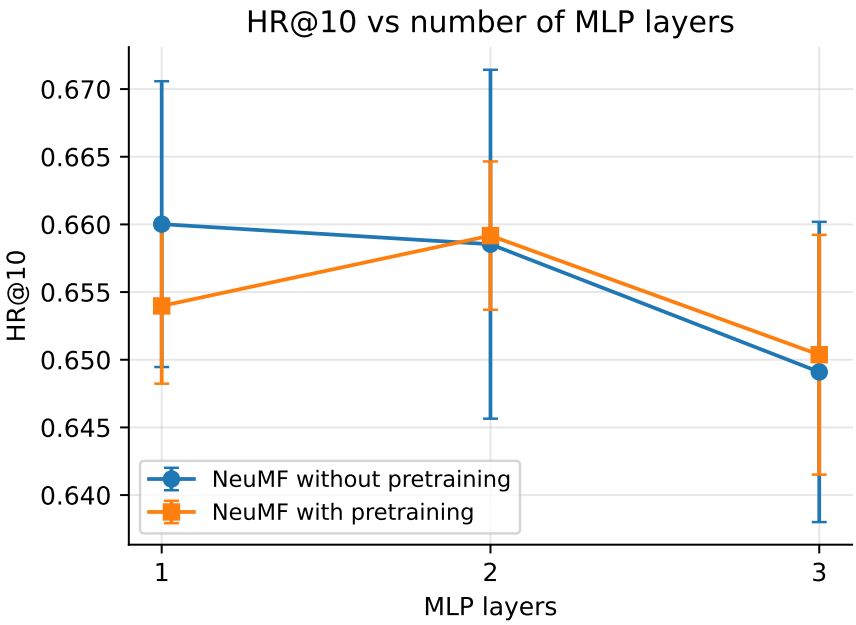


Figure 1: HR@10 of NeuMF as a function of the number of MLP hidden layers, with and without pretraining on MovieLens-100K. Error bars are ±1 std over 10 runs.

Weight-parameter count vs. depth (Task 3). Table 2 and Figure 2 show how the number of trainable weight parameters of NeuMF scales with the MLP depth. The count grows very slowly (107,121 → 107,633 → 107,761, an increase of only 0.6 % from 1 to 3 layers) because the four embedding tables dominate the parameter budget: $7{,}544 + 13{,}456 + 30{,}176 + 53{,}824 = 105{,}000$ parameters in the embeddings alone, vs. at most $\sim 2{,}700$ parameters in the MLP weights themselves. In other words, *MLP depth is essentially free from a parameter-count perspective* — the real cost is in the embedding layers.

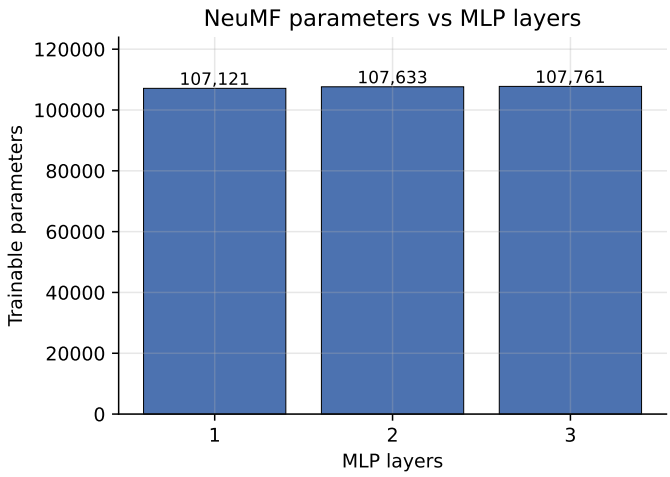


Figure 2: Trainable parameter count of NeuMF as a function of MLP depth. The embedding tables account for $\sim 97\,\%$ of the parameters regardless of MLP depth.

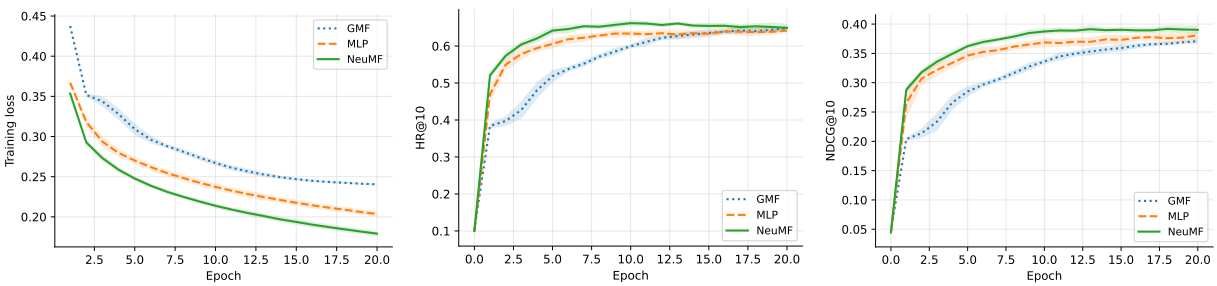
Table 2: NeuMF parameter count vs. MLP depth.

MLP layers	1	2	3
Trainable parameters	107,121	107,633	107,761

3 Training dynamics (Task 4)

Figure 3 shows the training loss, HR@10, and NDCG@10 of GMF, MLP, and NeuMF as a function of training epoch (shaded bands are ± 1 std over 10 seeds). This is the MovieLens-100K analogue of Figure 6 of [1]. Three observations stand out:

- **The ordering matches the paper's finding exactly.** Across all 20 epochs, NeuMF achieves the lowest training loss, followed by MLP, then GMF; and the recommendation-performance ordering (HR@10, NDCG@10) follows the same pattern in early epochs, which confirms that fusing linear (GMF) and non-linear (MLP) kernels helps model user–item interactions better than either alone.
- **The bulk of the improvement occurs in the first 5–10 epochs.** NeuMF reaches HR@10 ≈ 0.64 by epoch 5 and peaks at 0.662 around epoch 10, while its training loss continues to decrease monotonically all the way down to ~ 0.18 by epoch 20. This is a textbook pattern of mild overfitting: the training objective keeps improving even after the test-time metric has plateaued, just as the paper reports. After epoch ~ 13 the test HR@10 fluctuates slightly below its peak.
- **By epoch 20 the three models nearly converge at HR@10.** GMF, MLP and NeuMF all end up within 0.01 of each other in HR@10 (~ 0.65) even though their training losses remain very different. This suggests that, on this small dataset, the difficulty of the task (ranking one positive among 100 items) is the binding constraint rather than the expressive capacity of the interaction function.



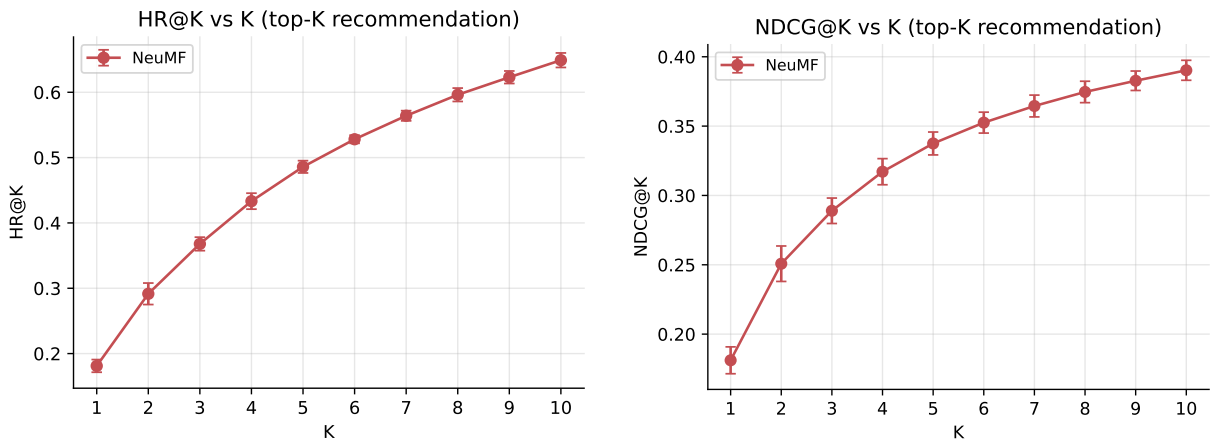
(a) Training loss (b) HR@10 (c) NDCG@10

Figure 3: Per-epoch training dynamics on MovieLens-100K. Shaded bands are ± 1 std over 10 seeds.

4 Top-K recommendation performance (Tasks 5 & 6)

Figure 4 shows how NeuMF's HR@K and NDCG@K vary with the ranking cutoff K from 1 to 10. Both curves grow monotonically with K: HR@1 is 0.181 ± 0.010 (i.e. the model places the correct item first for $\sim 18\,\%$ of users on a list of 100 candidates) and grows to HR@10 $= 0.649 \pm 0.011$. NDCG@K grows more slowly ($0.181 \rightarrow 0.390$) because its logarithmic position discount de-emphasises positions deep in the list. The HR@K curve is roughly concave, which is

typical: the largest marginal gains come from small values of K (every additional slot at the top is more valuable than the next), after which the curve levels off.



(a) HR@K (b) NDCG@K

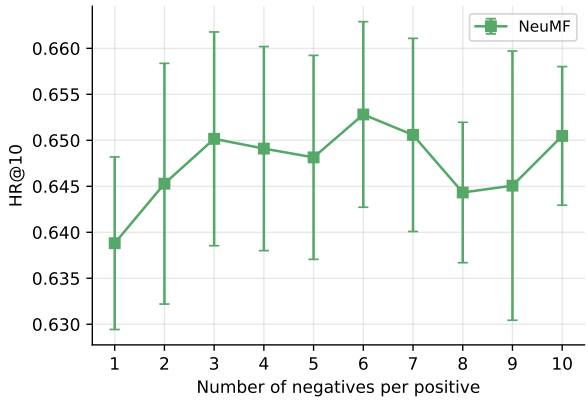
Figure 4: NeuMF top-K recommendation performance on MovieLens-100K, $K = 1, \ldots, 10$. Error bars are ± 1 std over 10 seeds.

5 Effect of the negative-sampling ratio (Tasks 7 & 8)

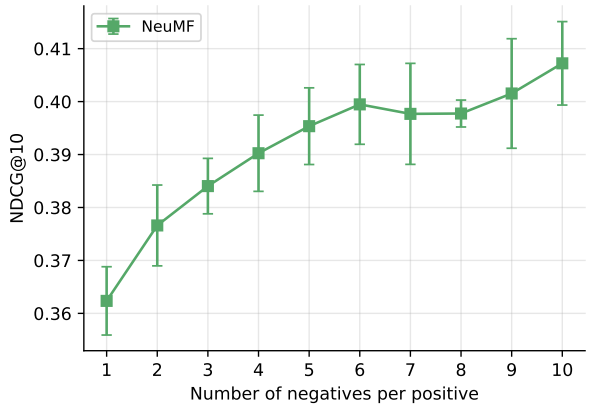
Figure 5 shows the effect of the number of negative samples drawn per observed positive at training time on the test-set HR@10 and NDCG@10. Two patterns emerge:

- **HR@10 rises sharply from 1 to 3 negatives** (0.639 $\rightarrow$ 0.650), then fluctuates in a narrow band (0.644–0.653) up to 10 negatives. The peak is at 6 negatives with HR@10 = 0.653 ± 0.010, but the plateau is broad and the differences beyond 3 negatives are well within one standard deviation of each other.
- **NDCG@10 grows more monotonically and does not plateau in our range.** It rises from 0.362 at 1 negative to 0.407 at 10, a $\sim 12\,\%$ relative improvement. The finer-grained nature of NDCG (it rewards putting the positive *higher* within the top-10, not just anywhere in the top-10) evidently benefits more from the additional negative signal, because larger negative pools make the contrastive objective sharper.

Both trends are qualitatively consistent with Figure 7 of the paper, which also finds that a single negative per positive is insufficient and that 3–6 negatives are enough to saturate HR@10. Our 10-negatives peak for NDCG is slightly higher in relative terms than the paper's (possibly because MovieLens-100K is smaller and benefits more from additional contrastive pressure), but the overall shape is the same.



(a) HR@10



(b) NDCG@10

Figure 5: Effect of the number of negative samples per positive interaction on NeuMF. HR@10 saturates after ~ 3 negatives; NDCG@10 keeps improving up to 10.

6 Non-negative matrix factorisation (Tasks 9 & 10)

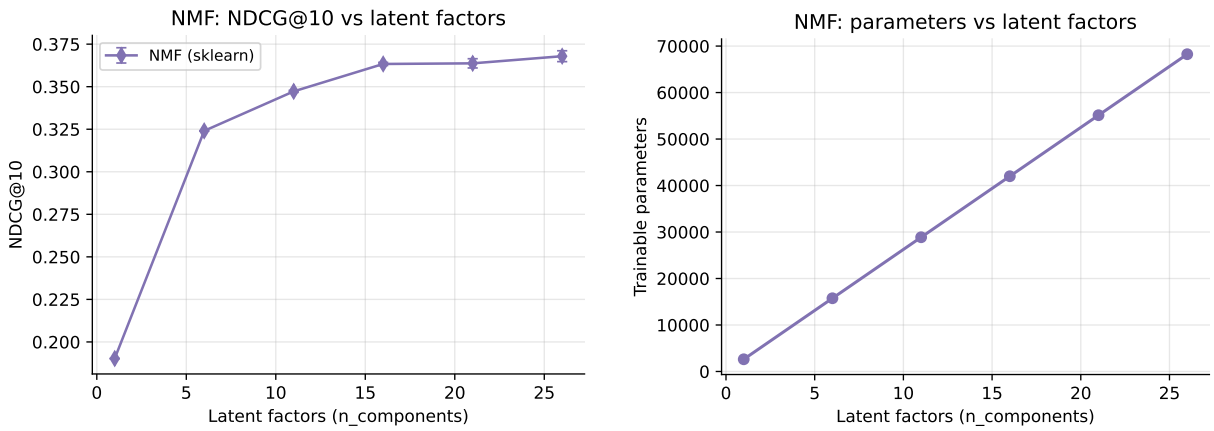
We build a binary user×item interaction matrix $R \in \{0,1\}^{943\times 1682}$ from the training positives only and factor it with `sklearn.decomposition.NMF` into $R \approx WH$, where $W \in \mathbb{R}_{\geq 0}^{943\times k}$ and $H \in \mathbb{R}_{\geq 0}^{k\times 1682}$. The reconstructed matrix $\hat{R} = WH$ gives a score $\hat{R}_{ui}$ for every user–item pair, which we then feed into exactly the same leave-one-out evaluation protocol as NeuMF (rank the held-out positive against the same 99 sampled negatives). The NMF model has $k \cdot (|U| + |I|) = 2{,}625k$ free parameters (the W and H matrices; we do not count transient internal state of the ALS iterations).

NDCG@10 vs. k (Task 9). Table 3 and Figure 6(a) show that NDCG@10 rises steeply between $k = 1$ and $k = 6$ ($0.190 \to 0.324$), then more slowly, plateauing around $k = 16$–26 at ~ 0.363–0.368. The best mean NDCG@10 is at $k = 26$ (0.368 ± 0.003), which we adopt as the "best NMF setting" for the rest of the report. The standard deviations are very small at small k (essentially zero for $k \in \{1, 6\}$, where the NNDSVD initialisation in scikit-learn is deterministic and the global optimum of a rank-1/6 factorisation is well defined), and grow modestly as k increases.

Parameter count vs. k (Task 10). Figure 6(b) shows the parameter count of NMF grows exactly linearly with k (it is $(943+1682)k$), from 2,625 at $k = 1$ to 68,250 at $k = 26$. In contrast to NeuMF — whose parameter count is dominated by the four fixed-size embedding tables and barely moves with MLP depth — NMF's parameter budget scales freely with the modelling capacity.

Table 3: NMF on MovieLens-100K: HR@10, NDCG@10 and parameter count vs. the number of latent factors k (mean ± std, 10 seeds).

Latent factors (k)	HR@10	NDCG@10	Parameters
1	0.3627 ± 0.0000	0.1902 ± 0.0000	2,625
6	0.5864 ± 0.0000	0.3240 ± 0.0000	15,750
11	0.6159 ± 0.0008	0.3472 ± 0.0006	28,875
16	0.6349 ± 0.0033	0.3633 ± 0.0009	42,000
21	0.6302 ± 0.0061	0.3637 ± 0.0027	55,125
26	0.6349 ± 0.0046	0.3679 ± 0.0032	68,250



(a) Task 9: NDCG@10 vs. k

(b) Task 10: parameter count vs. k

Figure 6: NMF sweep over the number of latent factors, $k \in \{1, 6, 11, 16, 21, 26\}$.

7 NeuMF vs. NMF comparison (Task 11)

Table 4 and Figure 7 compare the best NeuMF (3 MLP layers, no pretraining) with the best NMF ($k = 26$). NeuMF wins on both recommendation metrics: HR@10 is 0.649 ± 0.011 vs. 0.635 ± 0.004 for NMF (a relative improvement of $\sim 2.2\,\%$), and NDCG@10 is 0.390 ± 0.007 vs. 0.368 ± 0.003 ($\sim 6.1\,\%$ relative improvement). NeuMF's lead is more pronounced on NDCG@10 than on HR@10, which is consistent with the observations in Sections 4 and 6: NeuMF is especially good at pushing the correct item towards the top of the ranked list, and NDCG is the metric that most rewards that behaviour.

The comparison is less favourable to NeuMF once parameter count is considered: NeuMF uses 107,761 parameters against NMF's 68,250, a $57.9\,\%$ increase. On a per-parameter basis, NMF is actually a very strong baseline on MovieLens-100K — the small size of the dataset plays to the strengths of matrix factorisation, which has no under-parameterisation issues on a 943×1682 matrix with 26 latent factors. This observation motivates the knowledge-distillation experiments of the next section, which show that NeuMF's performance can be preserved with *fewer* parameters than NMF uses.

Table 4: Head-to-head comparison of NeuMF and NMF at their best respective settings (mean $\pm$ std, 10 seeds).

Method	HR@10	NDCG@10	Parameters
NeuMF (layers=3, no pretrain)	0.6491 ± 0.0105	0.3902 ± 0.0068	107,761
NMF (k=26)	0.6349 ± 0.0044	0.3679 ± 0.0030	68,250

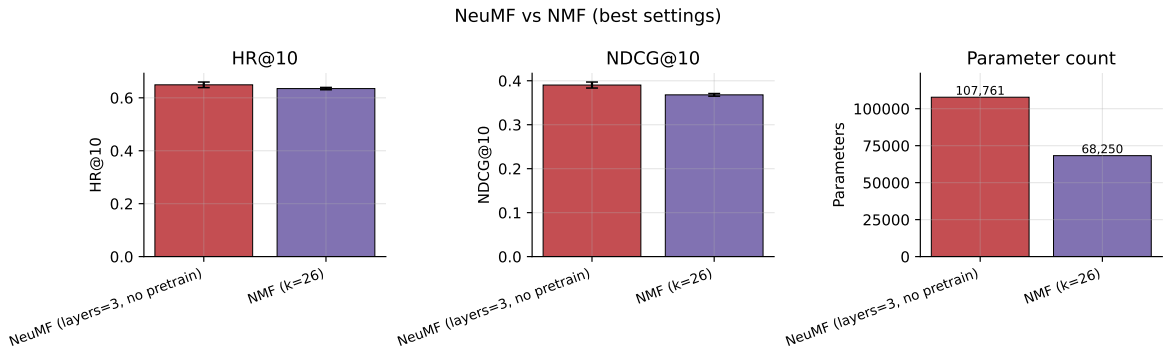


Figure 7: Best NeuMF (3 layers, no pretraining) vs. best NMF ($k = 26$) on HR@10, NDCG@10, and parameter count.

8 Knowledge distillation (Task 12)

We take the best NeuMF from the previous experiments as the teacher (107,761 parameters) and train three smaller NeuMF students, all with the same reduced architecture (`gmf_embed_dim`=4, `mlp_embed_dim`=16, 2 MLP layers, giving 53,177 parameters — a 50.7 % reduction). The three students differ only in the distillation objective, following the survey of Gou et al. [2]:

1. **Response-based KD** [3]: binary-cross-entropy between the student's temperature-scaled logits and the teacher's temperature-scaled soft probabilities. We combine with the standard hard-label BCE via an α-weighted sum ($\alpha = 0.5$, $T = 2$) and scale the soft term by T^2 to preserve gradient magnitudes as in the original paper.

2. **Feature-based KD / FitNets** [4]: MSE between the student's fused representation (concatenation of the GMF element-wise product and the MLP last hidden layer) and the teacher's fused representation. Because the two fused vectors have different dimensions (20 for the student, 72 for the teacher), we learn a linear projection $P \in \mathbb{R}^{72\times 20}$ of the student representation to the teacher's dimension. Total loss is hard-BCE $+\beta \cdot$ MSE with $\beta = 1$.

3. **Relation-based KD / RKD** [5]: within each mini-batch we compute the pairwise L_2-distance matrices of the (projected) student and teacher fused features, normalise each by its mean as in the original paper, and minimise the smooth-L_1 (Huber) loss between them. Total loss is hard-BCE $+\gamma \cdot$ RKD with $\gamma = 1$.

Table 5 and Figure 8 summarise the results. Three findings stand out:

- **Response-based KD matches and even slightly exceeds the teacher's HR@10** (0.655 ± 0.010 vs. 0.649 ± 0.011) while using **50.7 % fewer parameters**. NDCG@10 drops modestly from 0.390 to 0.378 (-3 % relative). This is the textbook outcome of knowledge distillation: the student inherits the teacher's ranking behaviour through the soft targets without needing the full architectural capacity to rediscover it from hard labels alone.

- **Relation-based KD retains most of the teacher's performance** (HR@10 0.630, NDCG@10 0.364), but is a few points behind response-based KD. Preserving pairwise-distance relations in representation space is evidently a weaker signal than directly copying the output distribution for this task.

- **Feature-based KD underperforms** (HR@10 0.582, NDCG@10 0.330, $\sim$10 % and $\sim$15 % relative drops respectively). Forcing the student's 20-dimensional fused representation to linearly project onto the teacher's 72-dimensional one is a strong constraint that may be

over-regularising the student and diverting gradient budget away from the actual ranking objective. FitNets-style hint losses typically benefit from careful layer selection and β-tuning, neither of which we explored here.

The practical takeaway is that on MovieLens-100K a NeuMF student with only 53,177 parameters — *fewer than the 68,250 used by the best NMF baseline* — can preserve the teacher's HR@10 when trained with response-based distillation. In other words, the parameter-count disadvantage of NeuMF observed in Section 7 is largely closeable through KD.

Table 5: Teacher NeuMF vs. three distilled students with the same smaller architecture (mean ± std, 10 seeds). All three students have the same parameter count; they differ only in the distillation loss used during training.

Model	HR@10	NDCG@10	Parameters	Reduction
Teacher (best NeuMF)	0.6491 ± 0.0111	0.3902 ± 0.0072	107,761	0.0%
Student 1: response-based KD	0.6550 ± 0.0098	0.3783 ± 0.0070	53,177	50.7%
Student 2: feature-based KD (FitNets)	0.5815 ± 0.0170	0.3302 ± 0.0118	53,177	50.7%
Student 3: relation-based KD (RKD)	0.6296 ± 0.0140	0.3643 ± 0.0079	53,177	50.7%

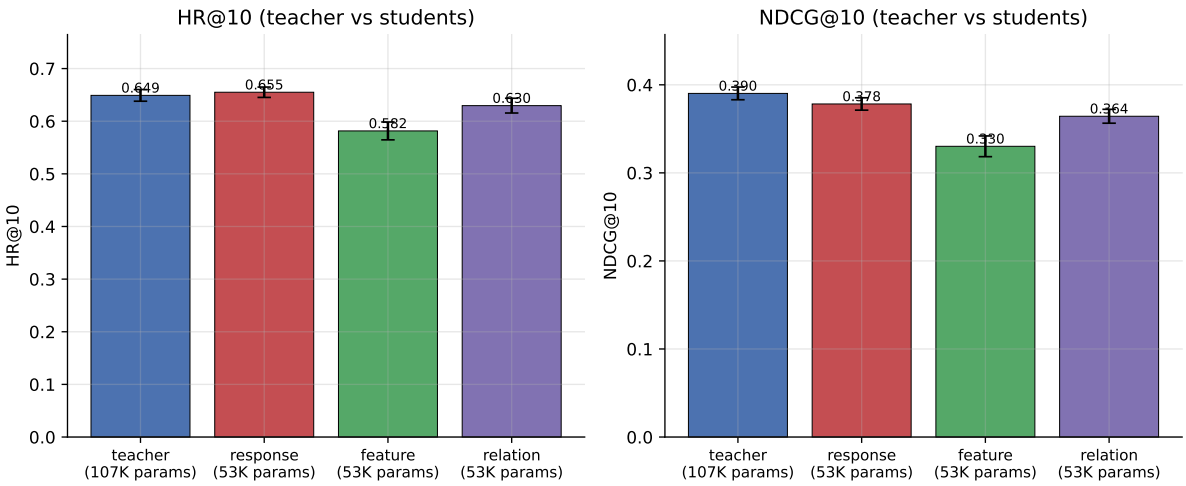


Figure 8: HR@10 and NDCG@10 of the teacher NeuMF (107,761 parameters) and three distilled students with identical reduced architecture (53,177 parameters each). Error bars are ±1 std over 10 seeds.

Reproducibility

All source code, experimental scripts, figures, tables and a Kaggle runner notebook are available at https://github.com/mavroul1s/ncf-assignment. The complete pipeline (10 seeds, 20 epochs per run, all 12 tasks) reproduces in roughly 3–4 hours on a single NVIDIA Tesla T4 GPU by running the provided Kaggle notebook; outputs are written to `results/`, which is consumed directly by this report.

References

[1] X. He, L. Liao, H. Zhang, L. Nie, X. Hu, and T.-S. Chua. *Neural Collaborative Filtering*. In WWW, 2017. arXiv:1708.05031.

[2] J. Gou, B. Yu, S. J. Maybank, and D. Tao. *Knowledge Distillation: A Survey.* International Journal of Computer Vision, 2021. arXiv:2006.05525.

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[5] W. Park, D. Kim, Y. Lu, and M. Cho. *Relational Knowledge Distillation.* In CVPR, 2019. arXiv:1904.05068.